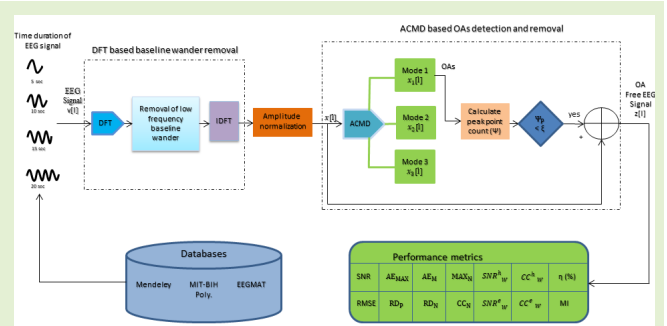


Automated Ocular Artifacts Removal Framework Based on Adaptive Chirp Mode Decomposition

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Abstract—The removal of ocular artifacts (OAs) from electroencephalogram (EEG) signals is crucial for effective and accurate analysis of the signals in different neurological and brain computer interface (BCI) applications. Towards this aspect, various multi-channel and single-channel techniques have been proposed for the removal/suppression of OAs from EEG signals. Recently, due to the rising demand of resource-limited ambulatory systems, single-channel techniques are more preferred for OAs removal. Although the existing techniques effectively remove OAs, they introduce distortion in the denoised EEG signal and the corresponding neurological features. Therefore, in this paper, we propose a framework for automated detection and removal of OAs from single-channel EEG signal based on discrete Fourier transform (DFT) and adaptive chirp mode decomposition (ACMD). The proposed framework is built in four stages: application of DFT to single-channel EEG signal; rejection of low-frequency baseline wander components; processed EEG signal decomposition using ACMD; rejection of mode containing OAs based on proposed peak point count threshold criteria. The performance of the proposed framework is assessed using noise-free EEG and EEG corrupted with OAs signals with different shapes and frequencies taken from three publicly available databases and in terms of different performance metrics. Further, comparative performance analysis demonstrates that the proposed framework outperforms the existing OAs removal techniques based on wavelet thresholding, variational mode decomposition (VMD), and Savitzky-Golay filter (SG-filter).

Index Terms—Adaptive chirp mode decomposition (ACMD), discrete Fourier transform (DFT), electroencephalogram (EEG), electro-oculogram (EOG), ocular artifacts (OAs).



I. INTRODUCTION

ELECTROENCEPHALOGRAPH (EEG) signal reflects the electrical activity of the brain, and it is a widely used modality to capture the brain dynamics [1]. It provides convenient, low cost, non-invasive, and portable access to the brain activity, among other advantages such as high temporal resolution, etc. [1]. To acquire EEG signals, various EEG devices such as NeuroSky, MindWave etc. are used as primary research tools for the study of neurological activities in these signals [2]. Also, these recorded signals are used to monitor several mental disorders like epilepsy, sleep, and anxiety disorders, etc. [3]. Therefore, accurate analysis of EEG signals is important for these neurological applications. However, the EEG signals are severely contaminated with different types of artifacts such as ocular artifacts (OAs), muscle artifacts, power line interference, etc. [4]. OAs are the most commonly

encountered artifacts during the acquisition of EEG signals, generated due to blinking of the eyes and movement of the eyeballs [5]. Presence of these artifacts causes difficulty in interpretation of EEG signals which further leads to improper diagnosis of mental disorders [6]. To avoid this, doctors usually advise the patients not to blink the eyes while recording EEG signals. However, this causes discomfort as they cannot stay still for long without blinking [6]. Therefore, artifact suppression/elimination is necessary rather than artifact avoidance. For this purpose, various researchers have proposed different multi-channel and single-channel approaches for the suppression/elimination of OAs, based on blind source separation (BSS), adaptive filtering, decomposition, and hybrid techniques. However, most of these techniques introduce distortion in the neurological features of EEG after the elimination of OAs [4]. Therefore, there is a need to develop a technique for the elimination of OAs while preserving the neurological features of EEG signals.

A. Related Work and Motivation

As mentioned above, different single-channel and multi-channel approaches have been proposed in the literary works for elimination of OAs, majorly based on BSS, adaptive filtering, decomposition, and hybrid techniques. Accordingly, we divide this sub-section into the different techniques with the corresponding literary works.

Manuscript received January 9, 2022; accepted January 25, 2022. Date of publication January 27, 2022; date of current version March 14, 2022. This work was supported by the Indian Council of Medical Research (ICMR) under Project ICMR/ISRM/12(89)/2020/2020-3701. The associate editor coordinating the review of this article and approving it for publication was Prof. Rosario Morello. (Corresponding author: Udit Satija.)

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Digital Object Identifier 10.1109/JSEN.2022.3147010

1) **BSS Techniques:** BSS techniques require multi-channel EEG signals for the suppression of OAs. Most of these techniques utilize independent component analysis (ICA) for separation of OAs from EEG signals. In [7], a combination of ICA and weighted support vector machine (SVM) has been proposed for the elimination of OAs based on different parameters such as ratio of the peak amplitude and variance, skewness, cross-correlation, Kullback Leibler divergence, and fluctuation of scalp distribution. In [8], Patidar and Zouridakis utilized iterative ICA and fuzzy clustering along with relative power and kurtosis features to eliminate OAs and muscle movement from the EEG signal. In [9], fastICA has been used to automatically reject OAs containing components without using any electro-oculogram (EOG) reference. In [10], a combination of K-means clustering and singular spectrum analysis (SSA) has been used to remove eye blinks from EEG signal without changing the real EEG signal.

2) **Decomposition Techniques:** Decomposition based techniques utilize mostly single-channel EEG signals for rejection of OAs by decomposing them into different components and are mostly based on empirical mode decomposition (EMD) and wavelet transform (WT). In [11], ensemble EMD (EEMD) and different features such as correlation, variance and skewness, have been exploited for the removal of OAs and cardiac artifacts. In [12], the blind deconvolution (BD) model has been investigated on the basis of complete EEMD with adaptive noise (CEEMDAN). In [13], an effective and robust framework based on variational mode decomposition (VMD) has been applied for the elimination of OAs from single-channel EEG signals. In [14], discrete WT (DWT) has been combined with a universal (DWT-UT) and statistical threshold (DWT-ST) for suppression of OAs from single-channel EEG signal.

3) **Adaptive Filtering Techniques:** Adaptive filtering has been widely used to remove OAs from EEG signals which needs a reference OA signal. In [15], two adaptive filters have been employed to remove OAs from EEG signal using a reference EOG. In [16], OAs have been removed from single-channel EEG signals using recursive least squares algorithm and adaptive noise canceller based filter. In [17], removal of OAs has been done using adaptive filtering and autoregressive-moving-average with exogenous inputs model modeling. In [18], adaptive filters based on ocular source components have been used to remove OAs from EEG signals. In [19], adaptive filtering has been applied to each of the recorded EEG signals with the help of the reference EOG for the removal of OAs.

4) **Hybrid Techniques:** Various hybrid techniques have also been used for OAs removal from single-channel EEG signals, and it requires the presence of reference OA/EOG signal. These techniques involve a combination of mostly BSS and decomposition based techniques. In [20], a moment matching approach for reducing stripe noise from hyper-spectral images has been adopted and refined to reduce OAs from EEG. Gang *et al.* in [6] proposed the utilization of multi-variate EMD (MEMD) for EEG signal decomposition and ICA for OAs and the signal component separation. Mammone *et al.* in [21] utilized kurtosis and Renyi's entropy characteristics obtained using the WT and ICA to remove several forms of artifacts such as linear trend, power line interference (PLI), electromyogram (EMG), and OAs. Castellanos and Makarov proposed the use of ICA for signal decomposition, wavelet thresholding for independent component demixing, and spectral coherence for OAs and cardiac artifact identification from the original EEG signal in [22]. Additionally, for automatic

elimination of OAs, Ghandeharion and Erfanian employed a mixture of ICA and wavelet using features such as mutual information, kurtosis, projection strength measure, and correlation coefficient in [23]. Burger and Heever used a wavelet-ICA based SVM classifier, as well as statistical characteristics such as kurtosis, variance, Shannon's entropy, and amplitude, to identify and remove eye blink artifacts in [24]. In [25], characteristics such as kurtosis, median, and average band power of EEG waves have been used by a hierarchical clustering-based automatic OAs removal approach. In [26], a digital filter has been used with decision rules to conduct artifact rejection. In [27], SSA and ICA have been used in conjunction with a stationary WT (SWT). In [28], a new hybrid technique based on EMD and ICA is presented to automatically remove the eye blink artifact from the EEG. In [29], removal of OAs using generative adversarial networks based deep learning model has been proposed.

Various techniques have been proposed, as aforementioned, there are some limitations which must be addressed. For example, the existing BSS techniques demand the presence of multi-channel EEG signals which restricts the use of these techniques in ambulatory environments. Furthermore, ICA based techniques assume the orthogonality condition between EEG signals and additive OAs, however due to the short time scales employed in real-time EEG analysis, the orthogonality of OAs and EEG can not be completely utilized [30]. Decomposition techniques such as EMD, suffers from the problem of mode-mixing [31], [32]. Although, VMD solves the problem of mode mixing by decomposing the modes at the same time, however, the number of modes have to be specified in advance and bandwidth parameter is fixed during the optimization. Adaptive filtering techniques require the reference EOG which may not always be available. Although many researchers have combined different techniques to form hybrid techniques, these techniques introduce distortion in the denoised EEG signal which leads to the loss of valuable signal information.

B. Objectives and Key Contributions

As mentioned above, the existing techniques such as BSS, decomposition, adaptive filtering and hybrid techniques perform OAs removal based on single and multi-channel EEG signals. Although the existing techniques are capable of removing OAs from EEG signals, there is a loss of valuable signal information from EEG signal after OAs removal. Most of the existing works directly apply denoising framework on input EEG signal whether the input EEG is OAs-free or OAs-contaminated EEG signal. This process distort the EEG signal even in case of OAs-free EEG signal. To overcome these limitations, in this paper, we propose a novel single-channel OAs detection and removal framework based on discrete Fourier transform (DFT), adaptive chirp mode decomposition (ACMD), and peak point count threshold criteria. The proposed framework involves four major stages: application of DFT to EEG signal; rejection of low-frequency baseline wander; processed EEG signal decomposition into modes using ACMD; rejection of mode containing OAs based on peak point count threshold criteria. The rest of this paper is organized as: Section II describes the major stages of the proposed framework. Section III depicts the performance evaluation results on the EEG signals taken from various databases for the removal of OAs. Lastly, Section IV concludes this work.

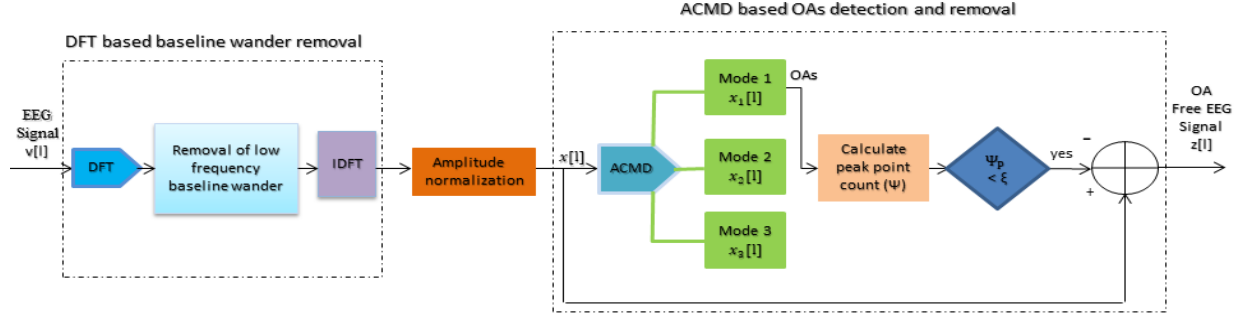


Fig. 1. Block diagram of the proposed ACMD-based OAs removal framework.

II. PROPOSED ACMD-BASED OAS REMOVAL FRAMEWORK

In this section, we describe the proposed framework for detection and removal of OAs from single-channel EEG signal, which is shown in Fig. 1. The proposed framework is built in four major stages: application of DFT to EEG signal; rejection of low-frequency baseline wander components; processed EEG signal decomposition into modes using ACMD; rejection of mode containing OAs based on proposed peak point count threshold criteria. These stages are described in the subsequent sub-sections.

A. DFT Based Baseline Wander Removal

In this work, we use a DFT based filtering technique to remove baseline wander [33], [34]. Let $v(l)$ be a discrete-time EEG signal with $l = 0, 1, 2, \dots, L-1$, where L denotes the number of samples. The DFT of $v(l)$ is calculated as follows:

$$V(k) = \sum_{l=0}^{L-1} v(l)e^{-j\frac{2\pi lk}{L}} \quad (1)$$

The baseline wander component is removed from the DFT coefficients with frequency values less than 1 Hz. The DFT coefficient index k for the component of F Hz is calculated as $k = \lceil \frac{FL}{F_s} \rceil$ with F_s as the sampling frequency. The baseline wander removed signal $u(l)$ is calculated as follows [35]:

$$u(l) = \frac{1}{L} \sum_{k=0}^{L-1} \tilde{V}(k)e^{j\frac{2\pi lk}{L}} \quad (2)$$

$\tilde{V}$ denotes the thresholded DFT coefficient vector which is obtained as $\tilde{V}(k) = [0, \dots, 0, V[k+1], \dots, V[L-k-1], 0, \dots, 0]$. Therefore, making coefficients zero in this frequency range after performing DFT gives the baseline wander component free signal $u(l)$. The extracted baseline wander signal $b(l)$ can be computed as:

$$b(l) = u(l) - v(l) \quad (3)$$

After the removal of the baseline wander components, the amplitude of signal $u(l)$ is normalized as:

$$x(l) = \frac{u(l)}{\max|u(l)|} \quad (4)$$

where, $x(l)$ denotes the normalized signal. Fig. 2 illustrates $b(l)$, $u(l)$ and $x(l)$ in (b), (c), (d), respectively, obtained from the original signal $v(l)$ in (a).

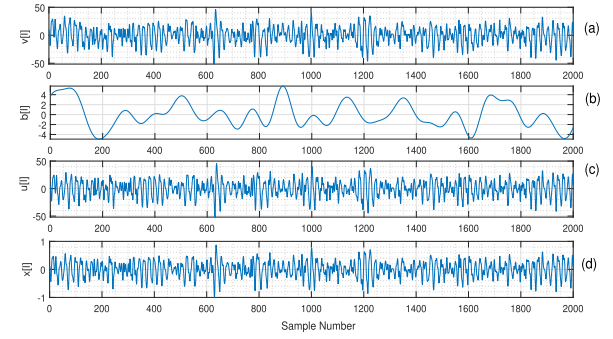


Fig. 2. (a) Original EEG signal taken from a subject of Mendeley database; (b) Extracted baseline wander; (c) Baseline wander free EEG signal; (d) Normalized EEG signal.

B. ACMD Based OAs Detection and Removal

1) **ACMD**: It is a recursive mode decomposition technique for decomposing a signal adaptively [36]. The key components of ACMD include adaptive bandwidth updating rule and instantaneous frequency (IF) (denoted as f_m) initialization scheme based on Hilbert transform [36]. ACMD utilizes a recursive mechanism to separate signal modes one by one using matching pursuit method. It resolves the problem encountered in the other decomposition techniques such as calculation of all the signal modes at the same time and instability. These problems can be solved by obtaining the m^{th} signal mode $x_m(l)$ after solving a minimization problem formulated as [36].

$$\min_{\vartheta_m, \varpi_m, \mathbf{f}_m} (\|\vartheta''\|_2^2 + \|\varpi''\|_2^2 + \tau \|x(l) - x_m(l)\|_2^2) \quad (5)$$

where,

$$x_m(l) = \vartheta_m(l) \cos(2\pi \int_0^l f'(s) ds) + \varpi_m(l) \sin(2\pi \int_0^l f'(s) ds) \quad (6)$$

where, $m = 1, 2, \dots, M$, M denotes the number of modes, $(\|\vartheta''\|_2^2)$ and $(\|\varpi''\|_2^2)$ are used for measuring bandwidth of the signal and for placing smoothness restrictions on the two de-chirped signals $\vartheta_m(l)$ and $\varpi_m(l)$ respectively. The term $(x(l) - x_m(l))$ denotes the residue of the signal ($r_{m+1}(l)$), $f'(s)$ is the estimated frequency of de-chirping factors $\cos(2\pi \int_0^l f'(s) ds)$ and $\sin(2\pi \int_0^l f'(s) ds)$, $\mathbf{f}_m = [f_m(l_0) \dots f_m(l_{L-1})]^T$ and τ represents the weighting factor

which is used for adjusting the weight of residual energy. The equation denotes a greedy algorithm, similar to that of the matching pursuit which finds the desired mode by lowering the residue's energy [36]. The signal is assumed to be observed in discrete time $l = l_0, l_1 \dots l_{L-1}$, where, L is the number of samples [36]. Substituting (4) into (5) yields the matrix form of the following objective function:

$$\mathbf{J}_l(\mathbf{y}_m, \mathbf{f}_m) = \|\mathbf{U}\mathbf{y}_m\|_2^2 + \tau \|\mathbf{x} - \mathbf{K}_m \mathbf{y}_m\|_2^2 \quad (7)$$

where, $\mathbf{x} = [x(l_0) \dots x(l_{L-1})]^T$, $\mathbf{y}_m = [\vartheta_m^T \varpi_m^T]^T$ with $\vartheta_m = [\vartheta_m(l_0) \dots \vartheta_m(l_{L-1})]^T$, $\varpi_m = [\varpi_m(l_0) \dots \varpi_m(l_{L-1})]^T$; Kernel matrix, $\mathbf{K}_m = [\mathbf{B}_m \mathbf{D}_m]$, where, $\mathbf{B}_m = \text{diag}[\cos(\psi_m(l_0)), \dots, \cos(\psi_m(l_{L-1}))]$, $\mathbf{D}_m = \text{diag}[\sin(\psi_m(l_0)), \dots, \sin(\psi_m(l_{L-1}))]$ with $\psi_m(l) = 2\pi \int_0^l f'(s)ds$;

$$\mathbf{U} = \begin{bmatrix} \mathbf{S} & \mathbf{0} \\ \mathbf{0} & \mathbf{S} \end{bmatrix} \quad (8)$$

where, $\mathbf{S}$ is a 2^{nd} order difference matrix with $(L-2) \times L$ given as:

$$\mathbf{S} = \begin{bmatrix} 1 & -2 & 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & -2 & 1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & 1 & -2 & 1 & 0 \\ 0 & \dots & 0 & 0 & 1 & -2 & 1 \end{bmatrix} \quad (9)$$

Algorithm 1 ACMD [36]

Input $\mathbf{x}$; parameters κ, τ^1 ; stopping thresholds ρ, ϵ
 $M \leftarrow \mathbf{r}_1 \leftarrow \mathbf{x}$
while $\|\mathbf{r}_m\|_2^2 / \|\mathbf{x}\|_2^2 > \rho$ **do**
 Set iteration counter $n = 0$, obtain the initial IF $f_m^1(l)$, and create matrix $\mathbf{K}_m^1$
 with $\mathbf{f}_m^1$
while $\|\mathbf{x}_m^n - \mathbf{x}_m^{n-1}\|_2^2 / \|\mathbf{x}_m^{n-1}\|_2^2 > \epsilon$ **do**
 $n \leftarrow n + 1$
 $\mathbf{y}_m^n = (\frac{1}{\tau} \mathbf{U}^T \mathbf{U} + (\mathbf{K}_m^n)^T (\mathbf{K}_m^n))^{-1} (\mathbf{K}_m^n)^T \mathbf{r}_m$
 $\mathbf{x}_m^n = \mathbf{K}_m^n \mathbf{y}_m^n$
 $\rho f_m^n(l) = \frac{\varpi_m^n (\vartheta_m^n(l))' - \vartheta_m^n (\varpi_m^n(l))'}{2\pi ((\vartheta_m^n(l))^2 + (\varpi_m^n(l))^2)}$
 $\mathbf{f}_m^{n+1} = \mathbf{f}_m^n + (1 + \frac{1}{\kappa} \mathbf{S}^T \mathbf{S})^{-1} \rho \mathbf{f}_m^n$
 Update kernel matrix $\mathbf{K}_m^{n+1}$
 $\tau^{n+1} = \frac{\tau^n (\mathbf{x}_m^n)^T \mathbf{x}_m^n}{(\mathbf{x}_m^n)^T \mathbf{r}_m}$
end while
 $\mathbf{x}_m = \mathbf{x}_m^n, \mathbf{f}_m = \mathbf{f}_m^{n+1}$ and residue $\mathbf{r}_{m+1} = \mathbf{x} - \mathbf{x}_m$
 $m \leftarrow m + 1$
end while
Output signal modes $\{\mathbf{x}_m\}_{m=1,2,3,\dots}$ and IFs $\{\mathbf{f}_m\}_{m=1,2,3,\dots}$

The other frameworks, like VMD, require the number of modes to be explicitly mentioned to get best accuracy however ACMD works on the recursive mechanism which extracts the desired chirp mode signal in the initial modes. The complete ACMD algorithm with the bandwidth adaption rule is outlined in Algorithm 1. The ACMD has two main tuning parameters i.e., τ and κ , where, τ controls the bandwidth of the modes and κ controls the instantaneous frequency of the modes. In this work, we have chosen values of these parameters such that most of the energy of ocular artifacts is captured in the first mode itself after experimental analysis. The experimental

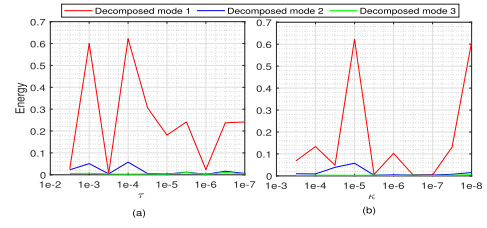


Fig. 3. Illustrates the energy for decomposed modes obtained from ACMD for different values of (a) τ ; (b) κ

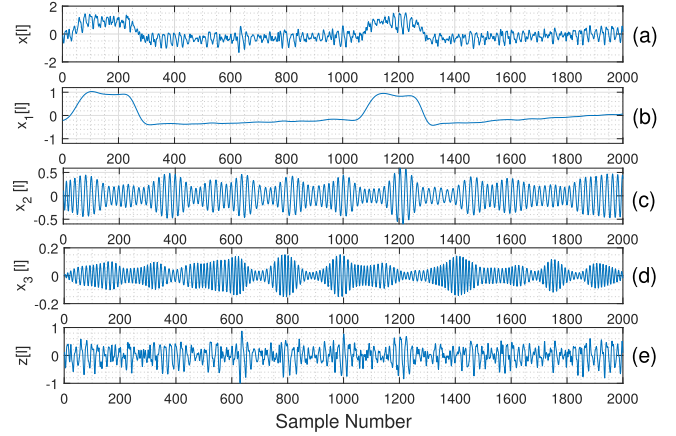


Fig. 4. (a) EEG signal contaminated by EOG signal taken from a subject of Mendeley database; (b), (c), (d) Decomposed modes using ACMD (first mode shows captured OAs); (e) Reconstructed EEG signal.

analysis results for κ and τ are presented in Fig. 3. It can be seen from Fig. 3 that for the value of τ and κ as $1e-5$ and $1e-6$, the maximum energy of OAs is captured in first decomposed mode. Therefore, we select these values of τ and κ in this work. Fig. 4 illustrates the decomposed modes obtained after applying ACMD to the signal $x(l)$, where, Fig. 4 (a) shows the contaminated signal and (b), (c), and (d) show the corresponding decomposed modes. It can be observed from the figure that first mode contains the OAs components.

2) Detection and Removal of OAs: In this work, we initially identify the presence of OAs in the first mode after ACMD and then reject this mode only if these are detected. To verify the presence of OAs in the first mode of ACMD, we have proposed peak point count based threshold criteria. The proposed peak point count (Ψ_P) is defined as:

$$\Psi_P = \sum_{p=1}^P h(p) \quad (10)$$

where, the peak of a signal $h(p)$ can be defined as:

$$h(p) = \begin{cases} 1 & |x_1(p-1)| < |x_1(p)| > |x_1(p+1)| \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

where $p = 1, 2, \dots, P$.

The value of Ψ_P for the first mode is depicted in Fig. 5 (a) for artifact-free and contaminated EEG signals taken from two databases. It can be observed from the figure that the peak point count values for OAs-free EEG signal are relatively higher than that for the contaminated EEG signal. Therefore, to discriminate OAs-free and OAs-contaminated signals, we apply a decision threshold (ξ) on Ψ_P . In this paper, the

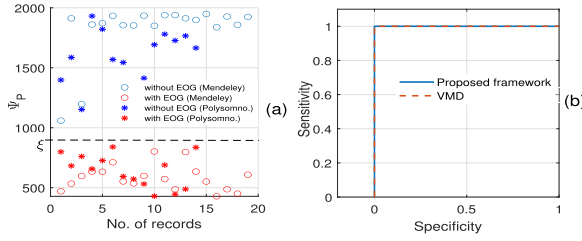


Fig. 5. (a) Illustrates the peak point count features in first mode of ACMD for the EEG signal with and without OAs taken from Mendeley database and MIT-BIH Polysomnographic database. (b) Illustrates the ROC curve for representing the comparative analysis of the proposed framework and existing techniques.

threshold is chosen as the middle value of the mean of peak point count in OAs-free EEG signal and mean of peak point count in OAs-contaminated EEG signal. If OAs are detected in the first mode, the reconstructed signal can be obtained as:

$$z(l) = x(l) - x_1(l) \quad (12)$$

where, $x_1(l)$ is the first mode after ACMD decomposition and $z(l)$ is the OA-free EEG signal. The proposed OA detector is summarized in Algorithm 2.

To compare the OAs detection performance of the proposed framework with existing techniques, we have computed the receiver operating characteristics (ROC) curve by plotting *Sensitivity* versus *Specificity* as shown in Fig. 5 (b). The

sensitivity (Se) can be calculated as: $Se = \frac{TP}{TP+FN}$, where, TP denotes the true positive (true contaminated EEG segments), FN denotes the false negative (missed contaminated EEG segments).

Algorithm 2 Proposed OA Detector

Require: ξ , Ψ_P

if $\Psi_P < \xi$ then

$z(l) = x(l) - x_1(l)$

else

$z(l) = x(l)$

end if;

III. RESULTS AND DISCUSSION

In this section, we present the performance evaluation results of the proposed framework on different EEG signals taken from three publicly available databases. We also describe the test databases in this section.

A. Test Databases

In this paper, we use EEG signals from three publicly available EEG databases to assess the performance of the proposed framework: semi-simulated Mendeley database [37], MIT-BIH Polysomnographic database [38], and EEG during mental arithmetic tasks (EEGMAT) database [38]. Mendeley database consists of pure and OA-contaminated EEG signals, and EOG signals. The recorded clean EEG signals and EOG signals are mixed to generate the OAs corrupted EEG signals since this database is primarily used to evaluate the effectiveness of several denoising frameworks. The database contains 54 records with EEG signals of 30 seconds duration, each with

19 channels according to 10-20 electrode placement system, and 54 records with different EOG signals. All the signals are sampled at a rate of 200 Hz. The MIT-BIH Polysomnographic database, on the other hand, consists of 18 records which contains over 80 hours of EEG and EOG recordings, as well as numerous other physiological signals, all recorded at a sampling frequency of 250 Hz. The EEGMAT database contains 60s EEG segments recorded from 36 subjects before and during the execution of mental arithmetic tasks, using a Neurocom EEG 23-channel device with electrode placement based on the 10-20 international system. The signals are sampled at 500 Hz. Since, EOG signals are present in the MIT-BIH Polysomnographic database records *slp32*, *slp37*, *slp41*, *slp45*, and *slp48*, we generate a large number of OAs-corrupted EEG signals by randomly mixing EOG signals with EEG signals from all records in these databases, after re-sampling all signals to a sampling rate of 200 Hz for accurate analysis. In the proposed framework, we have used all the channels of the three publicly available databases, giving each channel as a single input at a time.

B. Performance Metrics

In this sub-section, the performance assessment of the proposed framework for OAs removal is presented in terms of the following metrics: normalized cross-correlation (CC_N), signal to noise ratio (SNR), mean absolute error (AE_M), maximum absolute error (AE_{MAX}), normalized maximum absolute error (MAX_N), normalized root mean square difference (RD_N), percentage root mean square difference (RD_P), root mean square error ($RMSE$), weighted signal to noise ratio (SNR_W), weighted correlation coefficient (CC_W), percentage reduction in coefficient of correlation (η) [39], mutual information (MI) [40]. The metrics are calculated as [41]–[44]:

$$SNR = 10 \log_{10} \left[\frac{\sum_{l=1}^L (v(l) - \mu_0)^2}{\sum_{l=1}^L (v(l) - z(l))^2} \right];$$

$$AE_{MAX} = \max_{l=1}^L (|v(l) - z(l)|);$$

$$AE_M = \frac{\sum_{l=1}^L (|v(l) - z(l)|)}{L};$$

$$MAX_N = \frac{\max_{l=1}^L (|v(l) - z(l)|)}{\max(v(l)) - \min(v(l))};$$

$$RMSE = \sqrt{\sum_{l=1}^L (|v(l) - z(l)|)^2 / L};$$

$$RD_P = \frac{\sqrt{\sum_{l=1}^L (v(l) - z(l))^2}}{\sum_{l=1}^L (v(l) - 1024)^2};$$

$$RD_N = \frac{\sqrt{\sum_{l=1}^L (v(l) - z(l))^2 / L}}{\max(v(l)) - \min(v(l))};$$

$$CC_N = \frac{\frac{1}{L} \sum_{l=1}^L [v(l) - \mu_0] \sum_{l=1}^L [z(l) - \mu_r]}{\sqrt{\frac{1}{L} \sum_{l=1}^L [v(l) - \mu_0]^2} \sqrt{\frac{1}{L} \sum_{l=1}^L [z(l) - \mu_r]^2}};$$

$$\eta = \left[1 - \frac{1 - \rho_{vz}}{1 - \rho_{vx}} \right] \times 100;$$

$$MI = \sum_{i=1}^b \sum_{j=1}^b P_{(x,z)}(i, j) \log_2 \frac{P_{(x,z)}(i, j)}{P_x(i) P_z(j)}$$

where, L represents the number of samples in the EEG signal, $v(l)$ and $z(l)$ represent the original EEG and reconstructed signal respectively, μ_0 and μ_r are the mean of $v(l)$ and $z(l)$ respectively, $\rho_{vz} = \rho(v(l), z(l))$ and $\rho_{vx} = \rho(v(l), x(l))$ are the correlation coefficients calculated between original and reconstructed EEG signals, and original and contaminated EEG signals respectively, b is the number of bins, $P_{(x,z)}(i, j)$, $P_x(i)$ and $P_z(j)$ are the joint probability, probability of contaminated EEG signal and probability of reconstructed EEG signal. The mutual information (MI) parameter measures the distance between the probabilistic distribution of contaminated and reconstructed EEG signals. The high MI value denotes the better denoising performance of the proposed framework [44].

To measure the distortion in the neurological activity in each local rhythm ($\delta, \theta, \alpha, \beta, \gamma$), SNR_W and CC_W are presented in terms of performance metrics [43]. The SNR_W , based on relative wavelet subband energy (RWSEg) for d^{th} subband, can be calculated as:

$$SNR_W^e = E_d^{rel} \cdot 10 \log_{10} \frac{\sum_{l=1}^L |v_d(l)|^2}{\sum_{l=1}^L |r_d(l)|^2} \quad (13)$$

where, $r_d(l) = v_d(l) - z_d(l)$, $v_d(l)$ and $z_d(l)$ represent the original EEG and reconstructed EEG signal for d^{th} subband respectively, $E_d^{rel} = E_d/E^t$ with E_d as energy of the coefficients at the d^{th} level and E^t as the total energy of all the coefficients.

Similarly, SNR_W based on relative wavelet subband entropy (RWSEp) for d^{th} subband, can be calculated as:

$$SNR_W^h = H_d \cdot 10 \log_{10} \frac{\sum_{l=1}^L |v_d(l)|^2}{\sum_{l=1}^L |r_d(l)|^2} \quad (14)$$

where, $H_d = -E_d^{rel} \log(E_d^{rel})$.

The CC_W , based on RWSEg for d^{th} subband, can be calculated as:

$$CC_W^e = E_d^{rel} \cdot \frac{\sum_{l=1}^L |v_d(l) - \mu_{v_d}||z_d(l) - \mu_{z_d}|}{\sqrt{\sum_{l=1}^L |v_d(l) - \mu_{v_d}|^2 \sum_{l=1}^L |z_d(l) - \mu_{z_d}|^2}} \quad (15)$$

where, μ_{v_d} and μ_{z_d} are the mean of $v_d(l)$ and $z_d(l)$ respectively. Similarly, CC_W , based on RWSEp for d^{th} subband, can be calculated as:

$$CC_W^h = H_d \cdot \frac{\sum_{l=1}^L |v_d(l) - \mu_{v_d}||z_d(l) - \mu_{z_d}|}{\sqrt{\sum_{l=1}^L |v_d(l) - \mu_{v_d}|^2 \sum_{l=1}^L |z_d(l) - \mu_{z_d}|^2}} \quad (16)$$

Therefore, total measures SNR_W and CC_W , based on RWSEg and RWSEp, can be calculated as: $SNR_W^e = \sum_{d=1}^{D+1} SNR_W_d^e$; $SNR_W^h = \sum_{d=1}^{D+1} SNR_W_d^h$; $CC_W^e = \sum_{d=1}^{D+1} CC_W_d^e$; $CC_W^h = \sum_{d=1}^{D+1} CC_W_d^h$ where, D denotes the number of decomposition levels.

C. Performance Evaluation and Comparison

In this sub-section, we evaluate the proposed framework for 20 seconds duration of EEG signal and compare the performance with six existing techniques as shown in Table I. It can be observed from the table that the proposed framework

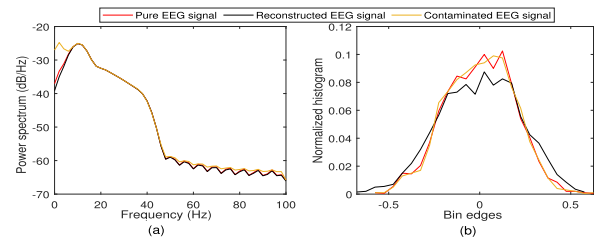


Fig. 6. (a) PSD plots for pure, contaminated and reconstructed EEG signals from a subject of Mendeley database. (b) Probabilistic distribution plots for pure, contaminated and reconstructed EEG signals from a subject of Mendeley database.

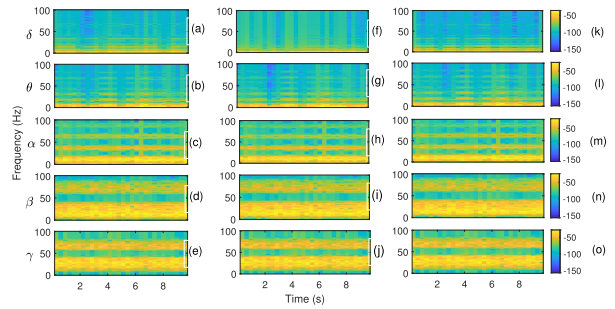


Fig. 7. Illustrates the short-time Fourier transform for the local rhythms ($\delta, \theta, \alpha, \beta, \gamma$) of the pure EEG signal in (a)-(e); Contaminated EEG signal in (f)-(j); Reconstructed EEG signal in (k)-(o) respectively.

outperforms the existing techniques in suppressing ocular artifacts in single-channel EEG signals for all three databases, in terms of all performance metrics. In this work, we analyze the power spectral density (PSD) and probabilistic distribution curves for the clean, contaminated and reconstructed EEG signals as shown in Fig. 6 (a)-(b). It can be observed that the presence of OAs yields higher power spectrum values for low frequency range, furthermore, the probabilistic distribution curve for the contaminated EEG signal yields lower probability values, indicating the presence of OAs. To illustrate the effectiveness of the proposed framework in preserving the signal information of EEG signal in its local rhythms ($\delta, \theta, \alpha, \beta, \gamma$), we have represented the short-time Fourier transform of different neurological activities in local rhythms of pure EEG signal in Fig. 7 (a)-(e), contaminated EEG signal in (f)-(j) and reconstructed EEG signal in (k)-(o) respectively. It can be observed from δ subband that energy of OAs is distributed in the contaminated EEG signal, whereas, after the removal of OAs, most of the energy of OAs is removed as obtained in the δ subband of the reconstructed EEG signal.

We also analyze the comparative denoising results for the existing techniques and proposed framework as illustrated in Fig. 8. It can be seen from the visual comparative analysis that existing techniques except VMD-based framework distort the EEG temporal characteristics and its clinical features. On the other hand, the proposed framework introduces minimum distortion while removing the OAs. We further analyze the comparative denoising performance in the EEG local rhythms i.e., $\delta, \theta, \alpha, \beta, \gamma$, which is shown in Fig. 9, where, it can be observed that the proposed framework outperforms the existing techniques in terms of different performance metrics in all three databases.

TABLE I
COMPARATIVE AVERAGE (STD) PERFORMANCE METRICS FOR ALL SUBJECTS OF DATABASES

Technique	Database	CC_N	SNR	AE_M	AE_{MAX}	MAX_N	RD_N	RD_P	RMSE	SNR_W^h	CC_W^h	SNR_W^o	CC_W^o	η (%)	MI
Proposed framework	Mendeley	0.997 (0.001)	22.990 (1.131)	0.0146 (0.004)	0.079 (0.034)	0.042 (0.017)	0.011 (0.002)	0.002 (3.72 e-04)	0.020 (0.004)	38.531 (3.653)	0.976 (0.023)	40.507 (4.063)	0.991 (0.015)	46.240 (8.850)	7.818 (0.142)
	MIT-BIH Poly.	0.995 (0.012)	20.282 (3.362)	0.016 (0.005)	0.107 (0.073)	0.061 (0.045)	0.013 (0.004)	0.002 (0.001)	0.023 (0.006)	36.024 (3.766)	0.935 (0.0326)	38.728 (9.435)	0.944 (0.056)	52.411 (9.434)	7.062 (0.492)
	EEGMAT	0.990 (0.032)	17.300 (2.559)	0.012 (0.025)	0.134 (0.236)	0.129 (0.125)	0.017 (0.015)	0.002 (0.003)	0.017 (0.030)	35.334 (5.533)	0.976 (0.0439)	37.928 (6.392)	0.986 (0.0533)	51.931 (12.981)	7.310 (0.306)
VMD [13]	Mendeley	0.995 (0.002)	18.030 (1.950)	0.026 (0.004)	0.116 (0.015)	0.063 (0.008)	0.017 (0.002)	0.003 (4.48 e-04)	0.031 (0.004)	26.787 (1.516)	0.952 (0.035)	29.041 (1.653)	0.983 (0.0234)	38.117 (8.291)	7.787 (0.105)
	MIT-BIH Poly.	0.976 (0.028)	15.480 (2.115)	0.029 (0.008)	0.215 (0.023)	0.117 (0.015)	0.022 (0.003)	0.004 (0.001)	0.039 (0.004)	27.691 (2.195)	0.966 (0.0132)	30.817 (3.560)	0.984 (0.008)	52.016 (8.208)	7.029 (0.591)
	EEGMAT	0.990 (2.61 e-05)	17.203 (2.834)	0.027 (4.22 e-04)	0.134 (0.004)	0.071 (0.007)	0.018 (0.001)	0.003 (5.13 e-05)	0.034 (5.24 e-04)	19.422 (2.882)	0.966 (0.022)	19.557 (4.343)	0.973 (0.0206)	42.063 (10.239)	7.300 (0.890)
DWT-UT [14]	Mendeley	0.905 (0.028)	7.080 (1.193)	0.079 (0.005)	0.603 (0.145)	0.324 (0.078)	0.057 (0.003)	0.010 (7.01 e-04)	0.105 (0.005)	21.06 (5.284)	0.88 (0.037)	19.90 (6.052)	0.90 (0.024)	37.874 (7.056)	7.703 (0.159)
	MIT-BIH Poly.	0.839 (0.045)	4.650 (1.477)	0.112 (0.008)	0.743 (0.108)	0.400 (0.057)	0.078 (0.005)	0.014 (0.001)	0.145 (0.010)	14.961 (3.840)	0.903 (0.026)	15.834 (4.244)	0.925 (0.029)	55.178 (6.749)	7.165 (0.485)
	EEGMAT	0.940 (0.017)	9.222 (9.090)	0.050 (0.002)	0.461 (0.094)	0.265 (0.147)	0.049 (0.014)	0.008 (9.12 e-04)	0.084 (0.009)	14.880 (2.686)	0.914 (0.054)	16.016 (2.965)	0.943 (0.064)	59.338 (14.616)	7.726 (0.104)
DWT-ST [14]	Mendeley	0.832 (0.041)	5.360 (1.149)	0.082 (0.016)	0.958 (0.080)	0.514 (0.039)	0.071 (0.013)	0.013 (0.002)	0.133 (0.025)	8.34 (3.789)	0.82 (0.042)	6.60 (3.561)	0.76 (0.048)	37.782 (13.355)	7.737 (0.110)
	MIT-BIH Poly.	0.742 (0.046)	3.050 (0.822)	0.117 (0.017)	0.993 (0.071)	0.534 (0.033)	0.089 (0.013)	0.016 (0.003)	0.167 (0.026)	8.502 (3.367)	0.883 (0.018)	8.260 (4.001)	0.894 (0.027)	45.629 (10.821)	7.476 (0.514)
	EEGMAT	0.885 (0.052)	7.063 (5.597)	0.062 (0.012)	0.829 (0.110)	0.478 (0.115)	0.065 (0.023)	0.011 (0.002)	0.113 (0.019)	10.348 (2.874)	0.905 (0.090)	8.394 (2.729)	0.939 (0.110)	60.858 (18.952)	7.778 (0.114)
SG-Filter [26]	Mendeley	0.920 (0.044)	8.080 (1.873)	0.052 (0.003)	0.570 (0.037)	0.360 (0.087)	0.052 (0.009)	0.008 (3.13 e-04)	0.085 (0.003)	14.57 (1.682)	0.89 (0.031)	13.20 (1.790)	0.87 (0.022)	35.454 (6.237)	7.716 (0.125)
	MIT-BIH Poly.	0.690 (0.124)	0.540 (3.479)	0.029 (0.008)	0.220 (0.033)	0.580 (0.283)	0.107 (0.044)	0.004 (0.001)	0.042 (0.008)	14.447 (3.040)	0.889 (0.068)	15.247 (4.073)	0.898 (0.0937)	52.479 (14.986)	7.146 (0.481)
	EEGMAT	0.959 (2.34 e-04)	11.183 (1.275)	0.044 (6.85 e-04)	0.453 (0.005)	0.241 (0.008)	0.036 (0.001)	0.006 (1.04 e-04)	0.068 (0.001)	17.711 (2.311)	0.860 (0.052)	7.293 (2.621)	0.840 (0.062)	28.082 (9.803)	7.156 (0.086)
EMD-ICA [28]	Mendeley	0.6551 (0.071)	2.2559 (0.797)	0.1775 (0.020)	0.763 (0.079)	0.4075 (0.025)	0.118 (0.009)	0.021 (0.003)	0.221 (0.027)	4.935 (0.347)	0.607 (0.075)	4.096 (0.681)	0.677 (0.113)	23.861 (4.76)	7.485 (0.024)
	MIT-BIH Poly.	0.783 (0.256)	4.125 (2.520)	0.117 (0.035)	0.515 (0.374)	0.294 (0.233)	0.085 (0.033)	0.014 (0.005)	0.149 (0.050)	5.531 (1.754)	0.71 (0.226)	12.143 (6.260)	0.684 (0.133)	51.874 (3.354)	7.609 (0.367)
	EEGMAT	0.55 (0.114)	0.847 (1.576)	0.227 (0.066)	0.889 (0.280)	0.467 (0.155)	0.151 (0.048)	0.028 (0.008)	0.287 (0.087)	6.068 (0.678)	0.597 (0.030)	2.217 (1.741)	0.430 (0.070)	38.181 (6.136)	7.323 (0.140)
MEMD-ICA [6]	Mendeley	0.931 (0.027)	8.515 (1.273)	0.0856 (0.009)	0.420 (0.123)	0.223 (0.062)	0.058 (0.006)	0.011 (0.001)	0.103 (0.013)	11.05 (0.872)	0.826 (0.027)	11.81 (1.102)	0.861 (0.048)	38.790 (8.994)	7.82 (0.118)
	MIT-BIH Poly.	0.847 (0.139)	5.478 (1.494)	0.101 (0.030)	0.4046 (0.232)	0.231 (0.425)	0.073 (0.021)	0.012 (0.004)	0.128 (0.041)	7.941 (1.551)	0.560 (0.049)	13.83 (3.873)	0.736 (0.118)	40.070 (11.663)	7.683 (0.604)
	EEGMAT	0.911 (0.065)	5.043 (1.423)	0.123 (0.025)	0.517 (0.125)	0.284 (0.096)	0.084 (0.015)	0.015 (0.003)	0.155 (0.031)	10.53 (1.922)	0.871 (0.043)	8.228 (3.543)	0.880 (0.095)	26.168 (9.064)	7.701 (0.153)

TABLE II
EFFECT OF PROCESSING EEG SEGMENT ON PERFORMANCE METRICS AND Ψ_P [MEAN (STD)] FOR THE PROPOSED FRAMEWORK

Time	Database	CC_N	SNR	AE_M	AE_{MAX}	MAX_N	RD_N	RD_P	RMSE	Ψ_P (EEG)	Ψ_P (OAs)	SNR_W^h	CC_W^h	SNR_W^o	CC_W^o	η (%)	MI
5s	Mendeley	0.996	21.473	0.024	0.109	0.061	0.018	0.003	0.032	486 (13.7)	111 (25.6)	31.5090 (6.292)	0.9396 (0.076)	35.0237 (7.141)	0.9719 (0.054)	38.5329 (13.235)	6.2089 (0.347)
	MIT-BIH Poly.	0.993	18.625	0.024	0.183	0.101	0.019	0.003	0.034	495 (101.6)	123 (52.4)	26.159 (9.051)	0.825 (0.133)	26.024 (12.881)	0.815 (0.165)	52.139 (13.996)	6.709 (0.122)
	EEGMAT	0.985	15.441	0.014	0.125	0.122	0.022	0.002	0.022	282 (149.5)	114 (33.2)	26.491 (9.798)	0.756 (0.048)	28.026 (11.91)	0.802 (0.143)	76.395 (27.075)	6.304 (0.309)
10s	Mendeley	0.996	21.350	0.019	0.078	0.041	0.013	0.002	0.024	963 (15.3)	200 (112.1)	32.6488 (8.457)	0.9726 (0.066)	33.6718 (7.574)	0.9818 (0.0409)	54.96 (8.702)	7.4297 (0.081)
	MIT-BIH Poly.	0.995	20.943	0.016	0.125	0.073	0.0132	0.002	0.022	988 (319.2)	331 (66.0)	27.431 (5.184)	0.876 (0.0629)	26.988 (12.235)	0.865 (0.140)	52.188 (18.495)	7.319 (0.0640)
	EEGMAT	0.990	17.461	0.012	0.113	0.109	0.016	0.002	0.017	852 (337.8)	183 (97.5)	23.987 (6.690)	0.788 (0.064)	27.697 (10.770)	0.851 (0.152)	68.08 (17.151)	6.534 (0.457)
15s	Mendeley	0.995	20.169	0.020	0.104	0.055	0.015	0.003	0.028	1300 (282.5)	318 (88.1)	36.1193 (4.009)	0.9421 (0.033)	39.267 (4.058)	0.9782 (0.019)	39.6949 (3.269)	7.7895 (0.087)
	MIT-BIH Poly.	0.996	20.485	0.017	0.140	0.082	0.013	0.002	0.023	1472 (417.7)	378 (93.0)	32.133 (5.802)	0.908 (0.0388)	32.658 (12.714)	0.895 (0.108)	49.346 (14.642)	7.248 (0.281)
	EEGMAT	0.992	18.021	0.011	0.066	0.063	0.015	0.002	0.016	1419 (681.5)	244 (195.9)	32.847 (10.328)	0.943 (0.064)	37.173 (13.968)	0.963 (0.107)	59.848 (19.725)	6.406 (0.675)
20s	Mendeley	0.997	22.990	0.0146	0.079	0.042	0.011	0.002	0.020	1948 (50.9)	551 (134.9)	38.531 (3.653)	0.976 (0.023)	40.507 (4.063)	0.991 (0.015)	46.240 (8.850)	7.818 (0.142)
	MIT-BIH Poly.	0.995	20.282	0.016	0.107	0.061	0.013	0.002	0.023	1692 (375.3)	429 (63.4)	36.024 (3.766)	0.935 (0.0326)	38.728 (9.435)	0.944 (0.056)	52.411 (9.434)	7.062 (0.492)
	EEGMAT	0.990	17.300	0.012	0.134	0.129	0.017	0.002	0.017	1604 (647.1)	391 (194.9)	35.334 (5.533)	0.976 (0.0439)	37.928 (6.392)	0.986 (0.0533)	51.931 (12.981)	7.310 (0.306)

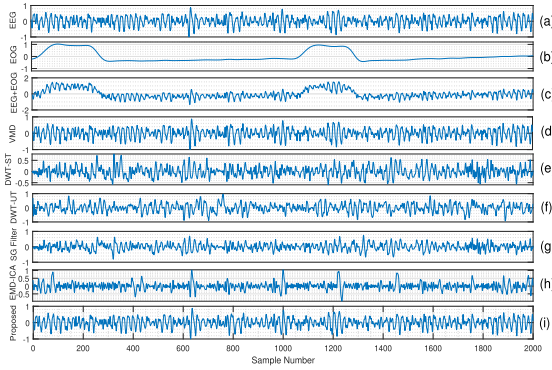


Fig. 8. Illustrates the comparative denoising results for the existing OAs removal techniques and proposed framework (a) Original artifact-free EEG signal taken from Mendeley database; (b) EOG signal taken from Mendeley database; (c) OAs-corrupted EEG signal; (d) Denoised signal using VMD [13]; (e) Denoised signal using DWT-ST [14]; (f) Denoised signal using DWT-UT [14]; (g) Denoised signal using SG filter [26]; (h) Denoised signal using EMD-ICA [28]; (i) Denoised signal using our proposed framework.

Since, the Mendeley database is recorded from the subject in sitting position, relaxed with eyes closed, therefore, it is most likely that the neurological activity is dominant in the α subbands [45], therefore, the proposed framework preserves the neurological information in the α subband better than the existing techniques. Similarly, the MIT-BIH Polysomnographic database is recorded during sleep, therefore, δ subband is most likely to be dominant during deep sleep and θ subband will be higher during light sleep. As compared to other techniques, our framework preserves the

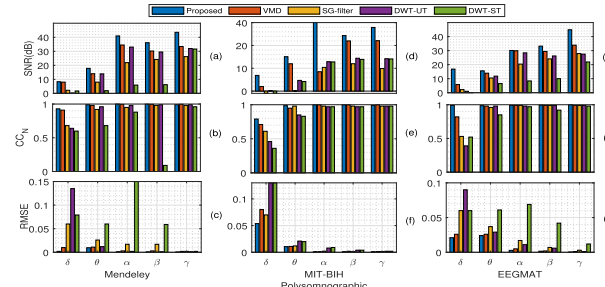


Fig. 9. Illustrates the bar charts for representing the comparative denoising performance of existing techniques and proposed framework in terms of three objective metrics in all local EEG rhythms For EEG signals taken from (a)-(c) Mendeley database; (d)-(f) MIT-BIH Polysomnographic database; (g)-(i) EEGMAT database.

neurological information in δ and θ subbands. Correspondingly, the EEGMAT database is recorded for the mental arithmetic tasks, therefore, the α subband is most likely to be dominant.

We also analyze the comparative performance of the proposed framework in terms of SNR , CC_N , $RMSE$, SNR_W^h , and MI for different channels of Mendeley and EEGMAT databases to demonstrate channel-wise denoising performance as shown in Fig. 10. It can be observed that frontal lobe channels (in particular, $F3$ and $Fp1$) outperforms other channels in terms of all the performance metrics for Mendeley and EEGMAT databases respectively. Furthermore, the comparative subject-wise performance of the proposed framework for MIT-BIH polysomnographic and EEGMAT databases is analyzed which

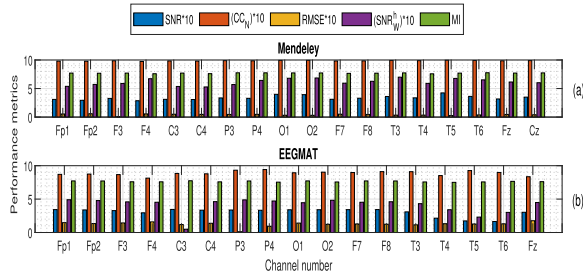


Fig. 10. Illustrates the bar chart for representing the comparative channel wise performance of proposed framework in terms of performance metrics for EEG signals taken from (a) 18 records of Mendeley database. (b) EEGMAT database.

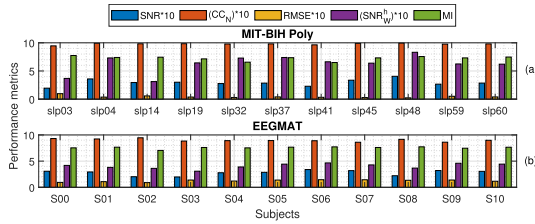


Fig. 11. Illustrates the bar chart for representing the comparative subject wise performance of proposed framework in terms of performance metrics for EEG signals taken from (a) 11 subjects of MIT-BIH Polysomnographic database. (b) 11 subjects of EEGMAT database.

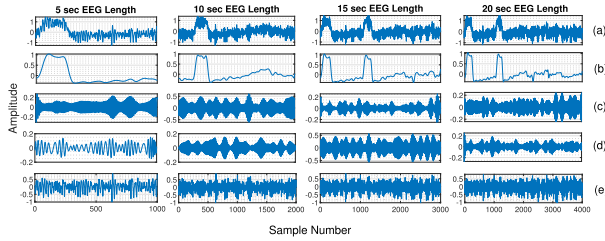


Fig. 12. (a) Original EEG signal taken from a subject of Mendeley database for 5, 10, 15 and 20 sec length segment respectively; (b)-(d) Decomposed modes; (e) Reconstructed EEG signals for 5, 10, 15 and 20 sec length segment.

is shown in Fig. 11. It is observed from Fig. 11 that subject 6 (S06) and slp45 of EEGMAT and MIT-BIH Polysomnographic databases respectively outperforms all the other subjects in terms of SNR , CC_N , $RMSE$, SNR_W , and MI . We also performed the computational complexity to evaluate the other single-channel and multi-channel OAs removal frameworks. The computation time for existing techniques like DWT-ST, DWT-UT, SG-filter, EMD-ICA, VMD, MEMD-ICA and proposed framework is 0.256 sec, 0.322 sec, 0.157 sec, 2.16 sec, 0.336 sec, 8.47 sec and 2.57 sec respectively. Although, the computational complexity of the proposed framework is higher as compared to existing techniques except multi-channel MEMD-ICA, the proposed framework outperforms other existing techniques in OAs removal without distorting EEG signal and does not require any information of channel correlation like multi-channel MEMD-ICA.

D. Impact of Input EEG Length

To further analyze the performance of the proposed framework on different EEG length segments, we have used 5, 10, 15, and 20 seconds duration of EEG signals. Firstly, we analyze the dependency of ACMD on decomposing different EEG

lengths. The decomposed modes using ACMD for different EEG lengths are illustrated in Fig. 12. It can be observed from the figure that ACMD is not length dependent. Then, we analyze the OAs removal performance for different EEG lengths. The corresponding performance metrics and peak point count (Ψ_P) for these length segments are presented in Table II. It can be observed from the table that processing a 20-second EEG segment produces a significant difference between the values of peak point count for OA-free EEG signal and OAs contaminated signal, which aids in determining a reliable threshold for OA detection. Furthermore, it provides superior OA rejection, as demonstrated by the performance metrics.

IV. CONCLUSION

In this paper, we proposed a framework for OAs removal from the single-channel EEG signal, based on DFT and ACMD. The proposed framework incorporates four stages: application of DFT to EEG signals; rejection of low-frequency baseline wander components; processed EEG signal decomposition into modes using ACMD; rejection of mode containing OAs based on the proposed peak point count threshold criteria. Evaluation results demonstrate that the proposed framework outperforms other existing techniques both in terms of subjective visual analysis and performance metrics tested on three public databases i.e., Mendeley, MIT-BIH Polysomnographic, and EEGMAT databases. Detailed experimental analysis demonstrates that ACMD is length independent, subject and channel independent in removing OAs.

ACKNOWLEDGMENT

The authors would like to thank Shrestha Walia for his help in some of the simulations.

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